

Lab # 05

The while loop

Objective

In this lab, we learned how to use loops in C++. The main focus was on:

- The **while loop**
- The **nested while loop**
- The **do...while loop**

Introduction

Loops are used in C++ to repeat a task multiple times. They help reduce code and make programs easier to write.

Types of Loops Studied

1. While Loop

- It checks the condition first.
- If the condition is true, the loop runs.
- If false, it stops.

2. Nested While Loop

- A while loop inside another while loop.
- Used when we need repetition inside another repetition.

3. Do...While Loop

- It runs the code at least one time.
- Then it checks the condition.

Conclusion

In this lab, we learned how different loops work in C++. We now understand how to repeat tasks using while, nested while, and do...while loops.

Lab 05 tasks

1: while Loop

- Write a C++ program that uses a while loop to display numbers from 1 to 10.
- Display each number on a separate line

Code:

```
#include <iostream>
using namespace std;

int main() {
    int i = 1; // yahan hum variable i ko 1 se start kar rahe hain

    while (i <= 10) { // jab tak i 10 se chhota ya barabar hai, loop chalega
        cout << i << endl; // har number ko alag line me print karega
        i++; // i ki value 1 se barhayi ja rahi hai
    }

    return 0; // program ka end
}
```

2. Repeat the first task using do... while loop

Code:

```
#include <iostream>
using namespace std;

int main() {
    int i = 1; // yahan hum variable i ko 1 se initialize kar rahe hain

    do {
        cout << i << endl; // pehle number print hoga
        i++; // phir i ki value increase hogi
    } while (i <= 10); // baad me condition check hogi

    return 0; // program ka end
}
```

3. Sum using while Loop

- Write a C++ program that prompts the user to enter a positive integer.
- Use a while loop to calculate the sum of all numbers from 1 to the entered integer.
- Display the calculated sum

Code:

```
#include <iostream>
using namespace std;

int main() {
    int n; // yeh variable user se number lene ke liye hai
    int i = 1; // counter variable 1 se start hoga
    int sum = 0; // sum ko 0 se initialize kiya gaya hai

    cout << "Enter a positive integer: "; // user ko input dene ke liye message
    cin >> n; // user se number lena

    while (i <= n) { // jab tak i n se chhota ya barabar hai
        sum = sum + i; // sum me i add kar rahe hain
        i++; // i ki value 1 se increase ho rahi hai
    }

    cout << "Sum = " << sum << endl; // final sum display karna

    return 0; // program ka end
}
```